

## ABURDEES, Egypt

WMO: 624580

Lat: 28.9099N

Lon: 33.1955E

Elev: 8

StdP: 101.23

Time Zone: 2.00 (E02)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 8.2        | 9.5 | -4.2                          | 2.7 | 15.8 | -2.1 | 3.2 | 16.0 | 9.1                   | 19.7 | 8.2 | 18.3 | 0.6                   | 40   | 0.401 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 6.2                    | 36.9            | 20.4 | 35.2 | 21.2 | 34.1 | 21.5 | 25.1                | 31.4 | 24.7 | 31.2 | 24.3 | 31.0 | 3.5                  | 300  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.0                            | 17.8 | 30.4 | 22.5 | 17.2 | 30.2 | 21.9 | 16.6 | 30.0 | 77.0          | 31.5 | 75.1 | 31.2 | 73.5 | 31.1 | 31.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n)  | (o)        | (p)  |      |     |
| 9.3               | 8.5  | 7.9 | DB                         | 5.0 | 42.4               | 2.0 | 1.7                                                | 3.6 | 43.6       | 2.4 | 44.6       | 1.3  | 45.6       | -0.1 | 46.8 | (4) |
|                   |      |     | WB                         | 1.9 | 27.3               | 1.1 | 2.3                                                | 1.1 | 28.9       | 0.5 | 30.3       | -0.1 | 31.5       | -0.9 | 33.2 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 23.6  | 15.3  | 16.5  | 19.6  | 22.8  | 26.2  | 28.7  | 30.0  | 30.5  | 28.8  | 25.6  | 21.3  | 17.2 | (6)  |
| (7)  |                                                                   | DBStd       | 5.75  | 2.19  | 2.62  | 2.98  | 3.04  | 2.66  | 1.94  | 1.45  | 1.32  | 1.62  | 2.03  | 2.41  | 2.22 | (7)  |
| (8)  |                                                                   | HDD10.0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                   | HDD18.3     | 232   | 99    | 63    | 17    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 49   | (9)  |
| (10) |                                                                   | CDD10.0     | 4957  | 164   | 183   | 299   | 383   | 503   | 561   | 621   | 635   | 564   | 483   | 339   | 222  | (10) |
| (11) |                                                                   | CDD18.3     | 2148  | 4     | 13    | 57    | 134   | 244   | 311   | 362   | 377   | 314   | 224   | 93    | 13   | (11) |
| (12) |                                                                   | CDH23.3     | 25503 | 21    | 77    | 366   | 1023  | 2524  | 3954  | 5068  | 5441  | 4152  | 2289  | 542   | 47   | (12) |
| (13) |                                                                   | CDH26.7     | 11439 | 2     | 13    | 116   | 333   | 967   | 1807  | 2631  | 2979  | 1879  | 629   | 80    | 2    | (13) |
| (14) | Wind                                                              | WSAvg       | 3.4   | 2.3   | 2.5   | 3.3   | 3.3   | 3.6   | 4.3   | 4.2   | 4.4   | 4.4   | 3.3   | 2.6   | 2.4  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 14    | 4     | 1     | 2     | 1     | 1     | 0     | 0     | 0     | 2     | 2     | 2     | 2    | (15) |
| (16) |                                                                   | PrecMax     | 54    | 33    | 6     | 14    | 9     | 15    | 0     | 2     | 0     | 1     | 11    | 25    | 12   | (16) |
| (17) |                                                                   | PrecMin     | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (17) |
| (18) |                                                                   | PrecStd     | 12    | 7     | 1     | 4     | 2     | 3     | 0     | 0     | 0     | 0     | 3     | 5     | 3    | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 26.0  | 28.6  | 33.7  | 36.6  | 39.4  | 39.3  | 39.0  | 37.8  | 36.7  | 34.3  | 31.2  | 26.6 | (19) |
| (20) |                                                                   | MCWB        | 16.7  | 16.9  | 18.2  | 18.2  | 19.3  | 19.7  | 22.0  | 21.3  | 21.2  | 20.9  | 19.2  | 17.6  | (20) |      |
| (21) |                                                                   | 2%          | DB    | 22.9  | 25.6  | 30.0  | 32.9  | 35.7  | 35.9  | 35.3  | 35.5  | 34.0  | 31.5  | 28.6  | 24.6 | (21) |
| (22) |                                                                   | MCWB        | 15.3  | 16.4  | 17.3  | 17.2  | 18.8  | 20.6  | 21.8  | 22.4  | 21.6  | 21.9  | 19.7  | 16.3  | (22) |      |
| (23) |                                                                   | 5%          | DB    | 21.3  | 23.5  | 27.0  | 30.3  | 33.0  | 34.1  | 34.1  | 34.3  | 32.8  | 30.3  | 27.3  | 23.2 | (23) |
| (24) |                                                                   | MCWB        | 14.6  | 15.1  | 16.0  | 17.3  | 18.9  | 20.8  | 22.1  | 22.8  | 22.2  | 22.1  | 19.3  | 15.6  | (24) |      |
| (25) |                                                                   | 10%         | DB    | 20.1  | 21.8  | 24.8  | 28.1  | 31.0  | 32.6  | 33.2  | 33.4  | 31.9  | 29.5  | 26.2  | 22.1 | (25) |
| (26) |                                                                   | MCWB        | 13.6  | 14.7  | 15.6  | 17.0  | 19.0  | 20.9  | 22.2  | 22.9  | 22.5  | 21.9  | 18.7  | 14.8  | (26) |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 18.4  | 19.7  | 20.7  | 20.7  | 22.2  | 24.0  | 25.4  | 26.1  | 25.5  | 25.0  | 22.4  | 19.5 | (27) |
| (28) |                                                                   | MCDWB       | 23.2  | 25.9  | 28.6  | 28.6  | 30.1  | 31.2  | 31.8  | 32.2  | 31.3  | 30.2  | 27.3  | 25.0  | (28) |      |
| (29) |                                                                   | 2%          | WB    | 16.6  | 17.6  | 18.7  | 19.7  | 21.5  | 23.2  | 24.7  | 25.2  | 24.8  | 24.1  | 21.5  | 17.9 | (29) |
| (30) |                                                                   | MCDWB       | 21.9  | 23.3  | 26.3  | 27.6  | 29.4  | 30.7  | 31.5  | 31.6  | 30.9  | 29.8  | 27.0  | 23.4  | (30) |      |
| (31) |                                                                   | 5%          | WB    | 15.2  | 16.3  | 17.5  | 18.9  | 20.9  | 22.7  | 24.2  | 24.8  | 24.3  | 23.4  | 20.5  | 16.5 | (31) |
| (32) |                                                                   | MCDWB       | 20.5  | 22.1  | 25.1  | 26.7  | 28.7  | 30.4  | 31.1  | 31.4  | 30.6  | 29.2  | 26.4  | 22.3  | (32) |      |
| (33) |                                                                   | 10%         | WB    | 14.0  | 15.2  | 16.5  | 18.2  | 20.4  | 22.2  | 23.7  | 24.5  | 23.8  | 22.8  | 19.3  | 15.3 | (33) |
| (34) |                                                                   | MCDWB       | 19.6  | 21.2  | 23.7  | 26.1  | 28.3  | 30.2  | 30.8  | 31.3  | 30.4  | 28.6  | 25.5  | 21.2  | (34) |      |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 9.0   | 9.1   | 8.9   | 9.0   | 8.6   | 7.8   | 6.8   | 6.2   | 6.0   | 6.8   | 8.6   | 9.1   | (35) |      |
| (36) |                                                                   | 5% DB       | MCDBR | 10.9  | 12.1  | 13.2  | 13.4  | 13.1  | 10.9  | 8.7   | 8.0   | 7.8   | 8.4   | 9.7   | 10.5 | (36) |
| (37) |                                                                   | MCWBR       | 7.7   | 7.7   | 7.0   | 6.0   | 5.1   | 5.0   | 4.7   | 4.7   | 5.2   | 6.1   | 6.9   | 7.4   | (37) |      |
| (38) |                                                                   | 5% WB       | MCDBR | 10.5  | 10.2  | 10.8  | 9.7   | 9.0   | 8.4   | 7.2   | 6.2   | 6.1   | 7.1   | 9.0   | 10.2 | (38) |
| (39) |                                                                   | MCWBR       | 8.3   | 7.5   | 6.7   | 5.4   | 4.6   | 4.6   | 4.4   | 4.4   | 4.9   | 5.9   | 7.6   | 8.4   | (39) |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.372 | 0.410 | 0.463 | 0.528 | 0.541 | 0.448 | 0.450 | 0.446 | 0.451 | 0.458 | 0.390 | 0.368 | (40) |      |
| (41) |                                                                   | taud        | 2.240 | 2.109 | 1.945 | 1.802 | 1.802 | 2.069 | 2.093 | 2.108 | 2.081 | 2.041 | 2.216 | 2.268 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 852   | 848   | 824   | 784   | 773   | 844   | 841   | 842   | 825   | 787   | 824   | 837   | (42) |      |
| (43) |                                                                   | Edh at Noon | 118   | 146   | 183   | 218   | 219   | 167   | 162   | 159   | 158   | 154   | 120   | 110   | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 3.96  | 4.92  | 6.28  | 7.30  | 7.98  | 8.51  | 8.27  | 7.70  | 6.73  | 5.44  | 4.33  | 3.70  | (44) |      |
| (45) |                                                                   | RadStd      | 0.15  | 0.18  | 0.19  | 0.17  | 0.20  | 0.09  | 0.10  | 0.09  | 0.13  | 0.14  | 0.12  | 0.10  | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (7 neighbors) | +0.40 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +135    | +146    |

Nomenclature: See separate page